

Chapter 6 Part II – Methods to estimate cost behavior

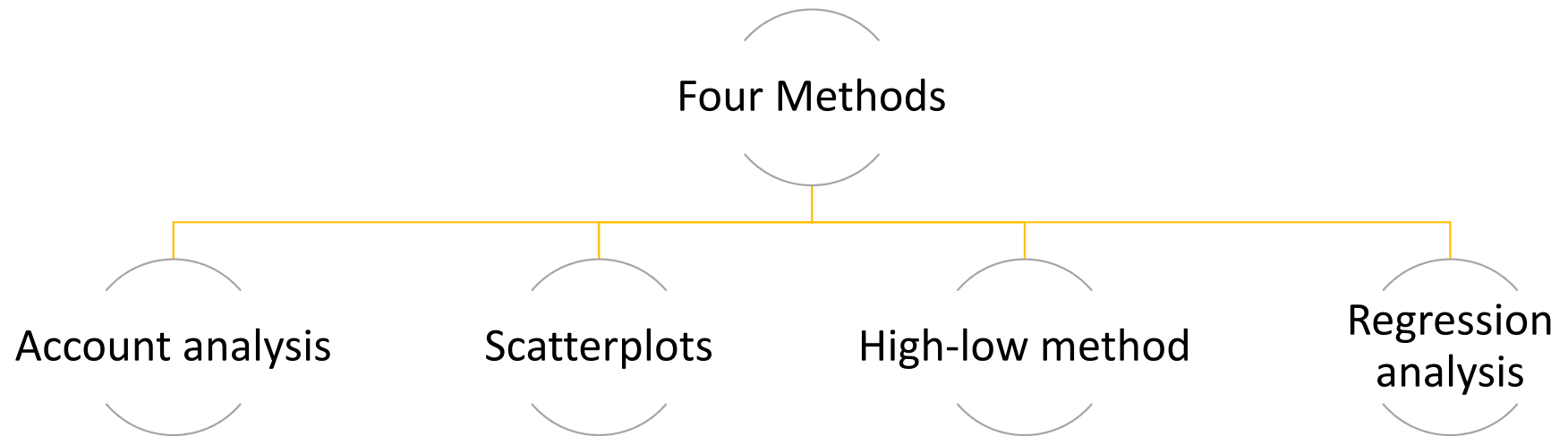
Learning Objectives

Use account analysis and scatterplots to analyze cost behavior

Use the high-low method to analyze cost behavior

Use regression analysis to analyze cost behavior

Analyzing Cost Behavior



Account Analysis

Using judgement to classify each general ledger account as variable, fixed, or mixed

Scatterplots

Graphs historical cost data on the y-axis and volume data on the x-axis



Helps visualize the relationship between cost and volume



Linear pattern → strong correlation

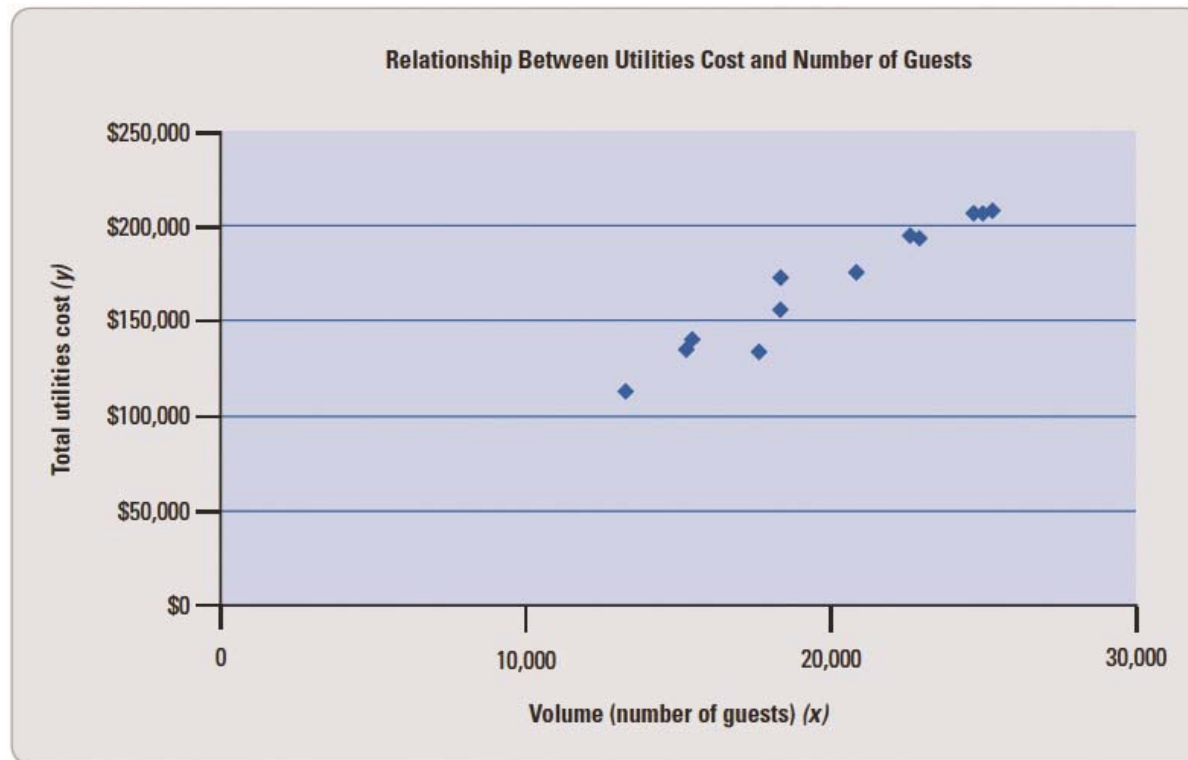


Random pattern → little to no correlation



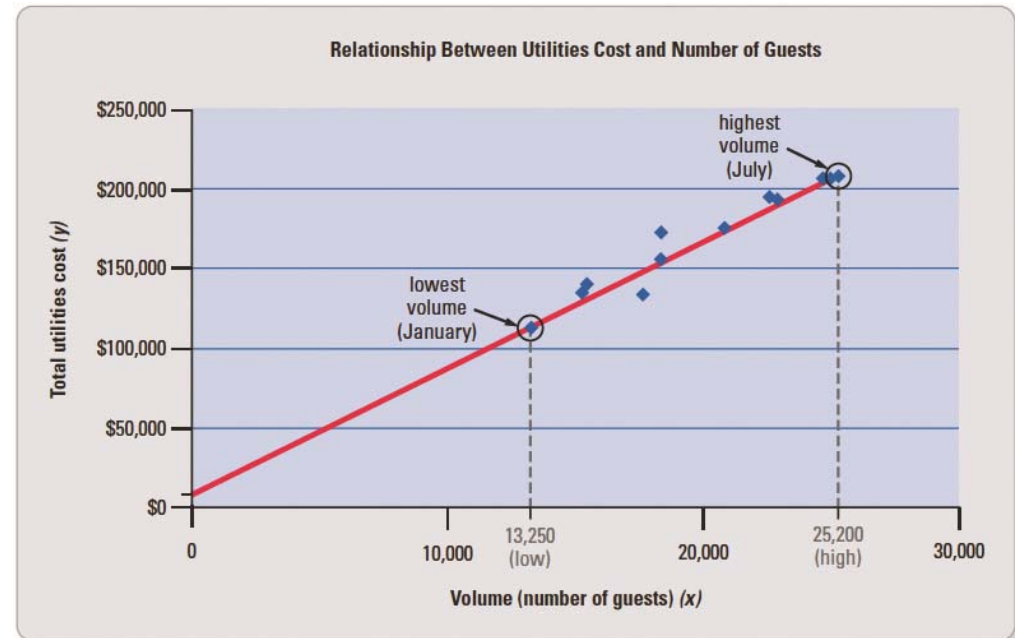
Outliers: abnormal data points

Scatterplot Graph



High-Low Method

- Fits a mixed cost line through the highest and lowest **volume** data points



High-Low Method – Step 1

Find the slope of the mixed cost line. The slope is the variable cost per unit of activity.

Equation of Mixed Cost Line: $Y = vX + F$;

Y = total cost; v = variable cost per unit; X = number of units, and F = fixed cost

To find slope:

Slope = Variable cost per unit of activity = Change in cost / Change in volume = $y(\text{high}) - y(\text{low}) / x(\text{high}) - x(\text{low})$

Example:

Units: Low—January 13,250; High—July 25,200

Costs: Low—January \$114,000; High—July \$209,600

$(\$209,600 - \$114,000) \div (25,200 - 13,250) = \8 per guest

High-Low Method – Step 2

Find the
vertical
intercept

the place where the line intersects the y-axis, the fixed cost component.

Use mixed
cost
equation:

$$y = vx + f$$

Insert either high or low data point and solve for f

$$\$209,600 = (\$8 \times 25,200) + f$$

$$f = \$8,000$$

Or

$$\$114,000 = (\$8 \times 13,250) + f$$

$$f = \$8,000$$

High-Low Method – Step 3

Using the amounts solved for in steps 1 and 2,

write the equation representing the costs' behavior.

Example:

$$y = \$8x + \$8,000, \text{ where}$$

y = total monthly utilities cost

x = number of guests

High-Low Method Conclusion

Advantages

- Simple
- Follows familiar $y=mx+b$

Disadvantage

- Only uses two data points

Regression Analysis



Statistical procedure for determining the line and cost equation that best fit all of the data points in the data set; <i>“the line of best fit”</i>
More accurate than high-low method
R-square: statistic tells us how well the line fits the data points; <i>“goodness-of-fit”</i>
Usually completed in Microsoft Excel or with a graphing calculator

Microsoft Excel Regression Analysis

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	<i>Regression Statistics</i>								
4	Multiple R		0.973273						
5	R Square		0.94726						
6	Adjusted R Square		0.941986						
7	Standard Error		8053.744						
8	Observations		12						
9									
10	ANOVA								
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
12	Regression	1	11650074512	1.17E + 10	179.6110363	1.02696E-07			
13	Residual	10	648627988.2	64862799					
14	Total	11	12298702500						
15									
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
17	Intercept	14538.05	11898.3624	1.221853	0.249783701	-11973.15763	41049.25	-11973.16	41049.25
18	X Variable 1	7.849766	0.585720166	13.4019	1.02696E-07	6.5446997	9.154831	6.5447	9.154831

R-Square

Higher the
R-square,

the stronger the relationship between cost and volume

Ranges

from 0 to 1

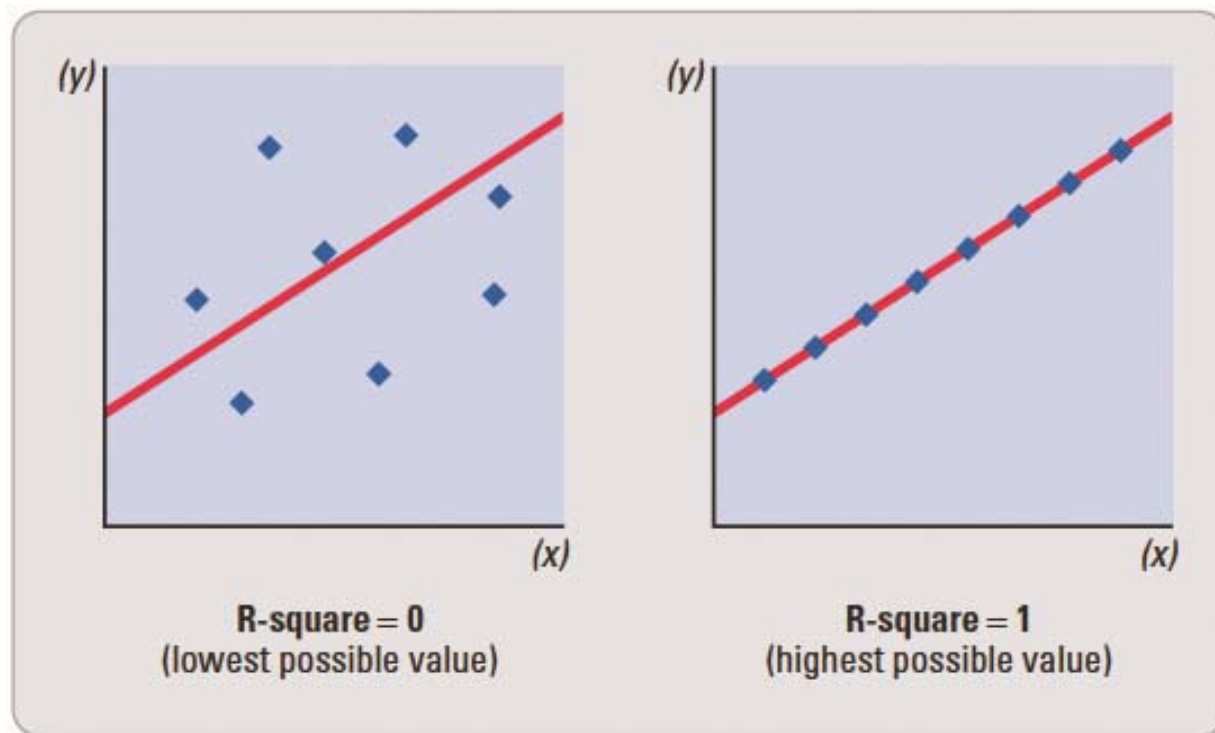
Rule of
thumb:

R-Square > 0.8 indicates very reliable cost equation

R-Square between 0.5 and 0.8 should be used with caution

R-Square < 0.5 indicates equation is not reliable

R-Square Graphs



Data Concerns - distortion

Only valid within
relevant range

Seasonality

Inflation

Outliers

May need to
remove from
data set

Summary of Linear Methods

Method	Approach	Advantages	Limitations
Visual Fit Method (Scattergraph)	Visual depiction of the relationship between X and Y.	Useful for getting a "feel" for the data.	Does not provide precise estimates of VC and FC
High-Low Method	Use the highest and lowest X value to fit the line.	Simple and intuitive.	Uses only 2 data points, which may not represent the general trend in the data.
Least-Squares Regression Method	Use a statistical package to find the line that minimizes the sum of squared error terms.	Uses all data points. Easy to calculate in Excel.	Requires more data and assumptions. Proper interpretation of results is critical.